



TNB RESEARCH
(199301004782)

Synthesis Report

DEVELOPMENT OF GREENHOUSE GAS (GHG) EMISSION FACTOR FOR THE ELECTRICITY SECTOR IN MALAYSIA

March 2021



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1 INTRODUCTION

TNB Research Sdn. Bhd. (TNBR) has embarked on a project to develop the Tier 2 country specific GHG emission factors for the electricity generation sector in Malaysia. The project was entitled Development of Greenhouse Gas (GHG) Emission Factor for the Electricity Sector in Malaysia and was awarded funding by the Akaun Amanah Industri Bekalan Elektrik (AAIBE), under Ministry of Energy, Science, Technology, Environment & Climate Change (MESTECC). The 36-months project aims to analyse and develop the major greenhouse gas (GHG) emission factor i.e. CO₂, CH₄ and N₂O from stationary combustion and establish a guideline to measure and report GHG emission data and a database to securely and systematically store GHG emission for the electricity sector in Malaysia. Four (4) main scopes were covered under this project. The main activities were the development of GHG emission factor which included sampling and data analysis, and the development of database as well as guideline.

MAIN FINDINGS

The main findings of this project was to provide the CO₂, CH₄ and N₂O emission factors for a period of three (3) years. The study period are 2017, 2018 and 2019. Additionally, the carbon content and net calorific value for each coal type and natural gas was also produced for national reference. The emission factors discussed are stipulated in **Table F.1** and **F.2**.

Table F.1: CO₂ Emission Factor for Coal and Natural Gas

Year	Fuel		CO ₂ [kgCO ₂ /TJ]	IPCC CO ₂ [kgCO ₂ /TJ]
2017	Coal	Bituminous	93,224	94,600
		Sub-bituminous	96,714	96,100
		Lignite	103,692	101,000
	Natural Gas		49,519	56,100
2018	Coal	Bituminous	93,078	94,600
		Sub-bituminous	96,679	96,100
		Lignite	101,720	101,000
	Natural Gas		50,023	56,100
2019	Coal	Bituminous	93,089	94,600
		Sub-bituminous	96,260	96,100
		Lignite	105,116	101,000
	Natural Gas		50,911	56,100

Table F.2: CH₄ and N₂O Emission Factor for Coal and Natural Gas

Fuel		CH ₄ [kgCH ₄ /TJ]	IPCC CH ₄ [kgCH ₄ /TJ]	N ₂ O [kgN ₂ O/TJ]	IPCC N ₂ O [kgN ₂ O/TJ]
Coal	Bituminous	0.10	1.0	0.70	1.5
	Sub-bituminous	0.09		0.95	
	Lignite	0.04		0.56	
Natural Gas		0.09	1.0	0.43	0.1

The weighted average for each year for CO₂ emission factor of Bituminous coal were lower than IPCC default values whilst Sub-Bituminous and Lignite coal are higher than the IPCC values. It was in the range of **93,078 kgCO₂/TJ to 93,224 kgCO₂/TJ**, for **bituminous coal** and **96,260 kgCO₂/TJ to 96,714 kgCO₂/TJ**, for **sub-bituminous coal**. The **lignite coal** showed higher value than the IPCC default which are within the range of **101,720 kgCO₂/TJ to 105,116 kgCO₂/TJ**. Overall, the net calorific value were higher for Bituminous 24.8 MJ/kg - 24.96 MJ/kg followed by Sub-Bituminous, 20.15 MJ/kg - 20.40 MJ/kg and Lignite, 15.05 MJ/kg - 15.66 MJ/kg whilst the

carbon content were within the range of 25.48 kg/GJ - 25.51 kg/GJ which was lower for Bituminous followed by Sub-Bituminous, 26.38 kg/GJ - 26.47 kg/GJ and significantly higher for Lignite coal, 27.85 kg/GJ - 28.81 kg/GJ. The natural gas emission factor was lower than IPCC value which is at **49,519 kgCO₂/TJ to 50,911 kgCO₂/TJ**. This is primarily due to the higher net calorific value of the natural gas consumed by the power plants by 9-12 % compared to IPCC default values.

The coal and natural gas CH₄ emission factor were lower from the Tier-1 fuel based method suggested by the IPCC. The CH₄ emission factor were 0.10 kgCH₄/TJ for Bituminous, 0.09 kgCH₄/TJ for Sub-Bituminous, 0.04 kgCH₄/TJ for Lignite and 0.09 kgCH₄/TJ for Natural Gas. Whereas the N₂O emission factor were lower for all coal types except for natural gas. The N₂O emission factor for Bituminous was 0.70 kgN₂O/TJ, Sub-Bituminous was 0.95 kgN₂O/TJ, Lignite coal was 0.56 kgN₂O/TJ and Natural Gas was 0.43 kgN₂O/TJ.

Below **Table F.3 – F.6** shows the average CO₂ emission factor, net calorific value, carbon content and coal oxidation factor.

Table F.3: Average National CO₂ emission factor for Coal and Natural Gas

Fuel		CO ₂ EF [kgCO ₂ /TJ]			
		National	Lower	Upper	Default IPCC
Coal	Bituminous	93,131	91,749	94,512	94,600
	Sub-bituminous	96,551	95,706	97,396	96,100
	Lignite	103,509	102,688	104,331	101,200
Natural Gas		50,151	46,365	55,993	56,100

Table F.4: Average Net Calorific Value for Coal and Natural Gas

Fuel		Net Calorific Value (MJ/kg)			
		National	Lower	Upper	Default IPCC
Coal	Bituminous	24.87	24.49	25.26	25.80
	Sub-bituminous	20.27	19.38	21.17	18.90
	Lignite	15.26	15.13	15.38	11.90
Natural Gas		52.95	46.57	56.70	52.22

Table F.5: Average Carbon Content for Coal and Natural Gas

Fuel		Carbon Content (kg/GJ)			
		National	Lower	Upper	Default IPCC
Coal	Bituminous	25.50	25.12	25.88	25.80
	Sub-bituminous	26.44	26.21	26.67	26.20
	Lignite	28.34	28.12	28.57	27.60
Natural Gas		12.59	12.42	13.02	15.30

Table F.6: Coal Oxidation Factor

Year	Oxidation Factor
2017	99.68
2018	99.61
2019	99.51

Additionally, the Guideline developed for this project documented existing methodologies and measurement tools which can be applied to power plants to continuously monitor and update their emission factor. Furthermore, the National GHG Emissions Inventory System was developed to enable power plants to comprehensively and reliably report inventories of their GHG emissions from stationary combustion. This system can be potentially used to develop new low-carbon

mitigation measures that can give power plant a competitive edge or pinpoint climate-related risks in its life cycle.

Through the adoption of these country specific emission factors to replace the default values, the GHG emissions assessed are more representative for the power plants. Through the assessment of GHG emissions for three consecutive years using country-specific emission factor, lower GHG emissions are observed by 1.5% for coal power plants and 11% for natural gas power plants thus translated to the GHG emissions amount which can be avoided by the country.

This project has exhibited that emission factor and emission amount changed due to changing carbon content as well as the calorific value. Therefore, Malaysia is suggested to continue to carry out this type of research in order to calculate and document a credible emission factor, because the calorific value and carbon content of coals and natural gas are changeable. The uncertainties also reported lower percentage for the country specific emission factor using the combined approach as shown in **Table F.7**.

Table F.7: Overall Uncertainty Assessment for Coal and Natural Gas

			Uncertainty (%)					
Fuel	Coal Type	Year	CO ₂	CH ₄	N ₂ O	IPCC CO ₂	IPCC CH ₄	IPCC N ₂ O
Coal	Bituminous	2017	1.88	22.29	60.13	7	50-150	Oder of Magnitude
		2018	1.33					
		2019	0.77					
	Sub Bituminous	2017	0.84	16.42	76.33			
		2018	0.92					
		2019	1.00					
	Lignite	2017	4.28	11.23	22.30			
		2018	3.96					
		2019	3.77					
Natural Gas		2017	1.00	17	54			
		2018	1.69					
		2019	1.82					

In addition, TNBR has established appropriate supporting database systems for the collection, aggregation and analysis of quantitative data for determination of these country-specific GHG emission factor for the stated period and boundaries.

2 METHODOLOGY

In this project, all coal power plants were sampled whereas natural gas power plants were selected. The natural gas power plant was selected based on factors such as accessibility, geographical location and approval process were considered in determining the suitability of the power plant for measurement activity. In order to meet the project objectives within the timeframe, this project was carried out at eight (8) CCGT power plants. The power plants were selected based on stratified sampling. The key objective is to conduct sampling with properly represented population. A stratified random sample is obtained by separating the population elements into non-overlapping groups, called strata, and then selecting a sample (often a simple random sample) independently from each stratum. To create a stratified random sample, there are six (6) steps: (a) defining the population; (b) choosing the relevant stratification; (c) listing the population according to the chosen stratification; (d) choosing your sample size; (e) calculating a proportionate stratification; and (f) using a simple random or systematic sample to select the sample.

A. Fuel Analysis

All the coal used by power plants in Peninsular Malaysia are imported while for Sarawak, local coal is used. The types of coal are being differentiated thus different emission factors are therefore analysed. In this assessment, a total of 1,949 coal samples were analysed. The analyses contain data on heat values and ultimate analysis (Carbon) needed for developing the relationship between carbon and heat content of the coal, that is, the carbon dioxide emission factors over the period of 2017, 2018 and 2019 in Peninsular Malaysia and Sarawak. The quantities of coal used in coal-fired power plants were sourced from a variety of brands. Analysis of CO₂ emission factor began with the data from Certificate of Analysis (COA) which sourced from TNB Fuel and data was taken from 2017, 2018 and 2019 for coal supplied to power generators according to each shipment. It includes the ultimate analysis of coal that quantify various elements present in the coal sample including carbon, hydrogen, sulphur, oxygen, and nitrogen contents. The elemental content of coal sample is usually presented in terms of percentage of mass fraction. The other important data is proximate analysis of coal which involves quantification of various element present in the coal sample including moisture, ash, volatile matter, and fixed carbon contents. The elemental content of coal sample usually presented in terms of percentage of mass fraction.

Actual coal data for each coal fired power plants are used which completeness check was conducted for each plant. However, some ultimate and proximate data were absence by Sarawak's plant. In this case, other sources of data from TNBR Combustion Lab was referred to for the missing values treatment. The data were then treated by statistical models.

Natural gas power plants in Malaysia utilises natural gas supplied by Petronas Gas Berhad. In Peninsular Malaysia, the natural gas is supplied through Peninsular Gas Utilization (PGU) Project pipeline, while Sabah and Sarawak make use of domestic pipelines from respective terminal.

The data was first given in daily or monthly basis and contains the natural composition percentage (%). Natural gas is composed mostly of methane (C_1), but also contains small amounts of ethane (C_2), propane (C_3), butane (C_4), pentane (C_5), hexane (C_6) and inert compounds (e.g., nitrogen, CO_2). Natural gas properties data such as heating value and density (MJ/kg) are also included in the set of data given. Other parameters that were also considered for development of GHG emission factor are gross generation (kWh) and total fuel consumption (GJ).

In the absence of parameters from the power plant, PETRONAS data was used to complete the data analysis. In addition, oxidation factor for the calculation of natural gas CO_2 emission factor is assumed to be 1. The 100% oxidation assumption is a common approach, adopted by the (IPCC, 2006), (API, 2009) and (IEA, 2019). All original data are converted into uniformed common units used in 2006 IPCC Guideline, such as "tC/TJ" in Net Calorific Value (NCV) from their original units such as "MJ/Sm³" in Gross Calorific Value (GCV).

B. Flue Gas Analysis

The emission of GHG for non- CO_2 was determined by TNBR during data collected from October 2018 to July 2019. Sample flue data were collected by using Gasmeter DX4000. The measurements involve identification of released greenhouse gas concentrations along with volumetric flowrate of the flue gas and heating value of the fuel at that time. The measurements activities were done using FTIR technique.

FTIR stands for Fourier Transform Infrared spectroscopy. It is a powerful gas measurement technology for simultaneous measurements of multiple gases. The ability to detect

and measure almost any gas, combined with the robustness and reliability of the technology, makes FTIR ideal for a wide variety of applications. All molecules can be identified by their characteristic absorption spectrum because each molecule absorbs infrared radiation at its characteristic frequencies. Every molecular structure has a unique combination of atoms, and therefore produces a unique spectrum when exposed to infrared light. Even molecules with the same number of atoms in different positions can be distinguished. The only exceptions are diatomic elements such as O₂ and noble gases, which do not absorb in the IR. IR absorption spectrum can be thought of as a fingerprint unique to each molecule.

The portable gas analyser used was Gasmet DX4000 as per **Figure 1** below for applications where multiple gas compounds need to be accurately monitored in hot & humid sample gas. The project has also developed and set up a sampling procedure for the direct measurement of the non-CO₂ gases.



Figure 1: GASMET DX4000 for Emission Gas Monitoring

From the data source, a total 3,109 samples for coal power plant and 2,687 samples for natural gas power plant was collected. Instead of carbon dioxide, flue gas emissions such as methane and nitrous oxide were taken to establish the emission factor. The data for CH₄ and N₂O emission was collected and statistical tools were applied for both flue gas datasets.

Secondary datasets are data obtained by applying certain formula, derived values and modelled data. During the cataloguing process, all data was reviewed and reported for

consistency, ambiguity of information, outliers etc. Data was also compared with default values reported by IPCC for assessing accuracy of secondary data obtained.

C. Greenhouse Gas Emission Factor

The derivation of GHG emission factor using fuel analysis data may vary significantly for different types of solid fuels, liquid fuels or gaseous fuels. In this project, emission factor equations for solid and gaseous fuels are sourced from the Korean National GHG Emission and Absorption Factor Development and Verification 1st Plan Report (Aug. 2014).

The overall GHG emission factor (GHG EF) framework for CO₂ is shown in **Figure 2**. The estimation of GHG emission factor adopted the fuel analysis method. In line with the guidelines provided by IPCC, the GHG EF for CO₂, will be discussed in the following section along with specific information and data on fuel.

Fuel analysis is one of the methods in deriving the GHG EF. This method utilises the properties of the fuel (carbon content and heating value) to determine the ratio of the amount of CO₂ to be released if the fuel is burned over associated heating value, thus obtaining the GHG EF. This section provides details on the fuel analysis for coal and natural gas to derive the GHG EF.

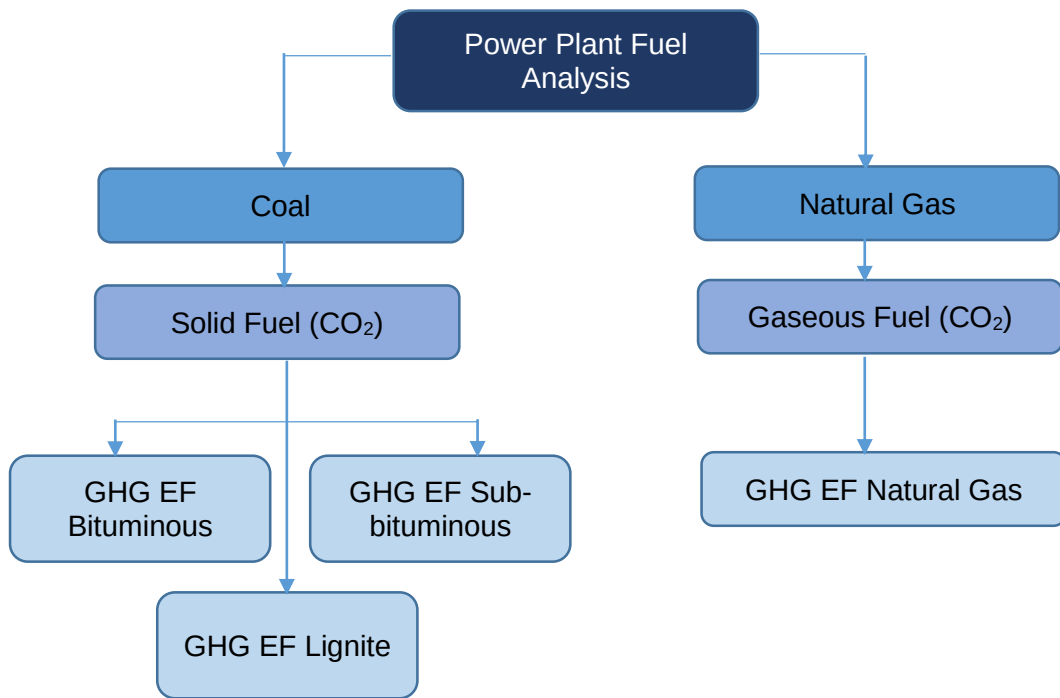


Figure 2: CO₂ Emission Factor Framework

- **Coal**

The equation below shows the derivation of GHG emission factor for coal. For solid fuel, the emission factor is involving the carbon content and net calorific.

$$EF_{i,CO_2} = C_{ar,i} / EC_i \times 3.664 \times 10^6 \times \text{Oxidation Factor}$$

Where;

- EF_{i,CO₂} : CO₂ emission factor for fuel (i) (kgCO₂/TJ-fuel)
- 3.664 : Molecular weight of CO₂ (44.010) / Atomic weight of Carbon (12.011)
- C_{ar,i} : Mass fraction of Carbon in fuel (i) (As received basis, between 0 and 1)
- EC_i : Calorimetric number of fuel (i) (Net fuel calorific value, MJ/kg-fuel)

The parameters of $C_{AR,i}$ and EC_i are required to be in as received basis. The oxidation factor calculated are based on below calculation (Fott, 2006)¹. The content of carbon in ash $C^{A}_{noncomb}$ (the so-called unburned carbon) has to be reduced to the total weight of the raw fuel $C^{r}_{noncomb}$

$$C^{r}_{noncomb} = A^r [C^{A}_{noncomb} / (1 - C^{A}_{noncomb})]$$

where A^r is the ash content in the raw fuel. As in the previous cases, the quality characteristics of the fuel express in the equations by weight fractions.

The oxidation factor is calculated as the ratio of the weight fractions of the burned and total carbon contained in the raw fuel:

$$f_{ox} = (C^{r}_{comb} / C^r) = 1 - (C^{r}_{noncomb} / C^r)$$

- **Natural Gas**

Greenhouse gas emission factor calculation for natural gas power plant is lengthier compared to coal power plants due to the nature of fuel supply composition as shown below.

$$EF_{i,CO_2} = \sum [(moly\% / V / \sum(moly\% \times mw_y / V)) \times (44.010 \times N_y \times f_i)] / EC_i \times D_i$$

Symbols used and their definitions and units are as follows:

EF_{i,CO_2} - CO₂ emission factor for fuel type i, [kgCO₂/GJ-fuel];

EC_i - calorific factor of fuel type i (net fuel calorific value), [GJ/1000m³-fuel];

D_i - density of fuel type i, [kg/1000m³-fuel];

$moly\%$ - composition of gas component y in 1 mol fuel type i, [%];

mw_y - molecular weight of gas component y of the fuel i, [kg/k-mol];

V - volume of 1 kmol of fuel type i under standard configuration (usually 22.4 m³), [m³];

N_y - number of carbon atoms in the gas component y of fuel type i, [-]; and

f_i - oxidation factor in fuel type i (1), [-]

¹ Fott et.al.2006, DETERMINATION OF EMISSION AND OXIDATION FACTORS IN MONITORING CO₂ EMISSIONS FOR EMISSION TRADING SCHEME, Acta Geodyn. Geomater., Vol. 3, No. 4 (144), 5-11, 2006.

Flue gas analysis is one of the methods in deriving the GHG EF. The overall framework is described in **Figure 3** below. This method involves direct measurement of the flue gas composition and properties of the fuel (heating value) to determine the ratio between amount of CO₂, CH₄, and N₂O to be released if the fuel is burned over associated heating value, thus the GHG EF. This section provides details on the flue gas analysis to derive the GHG EF.

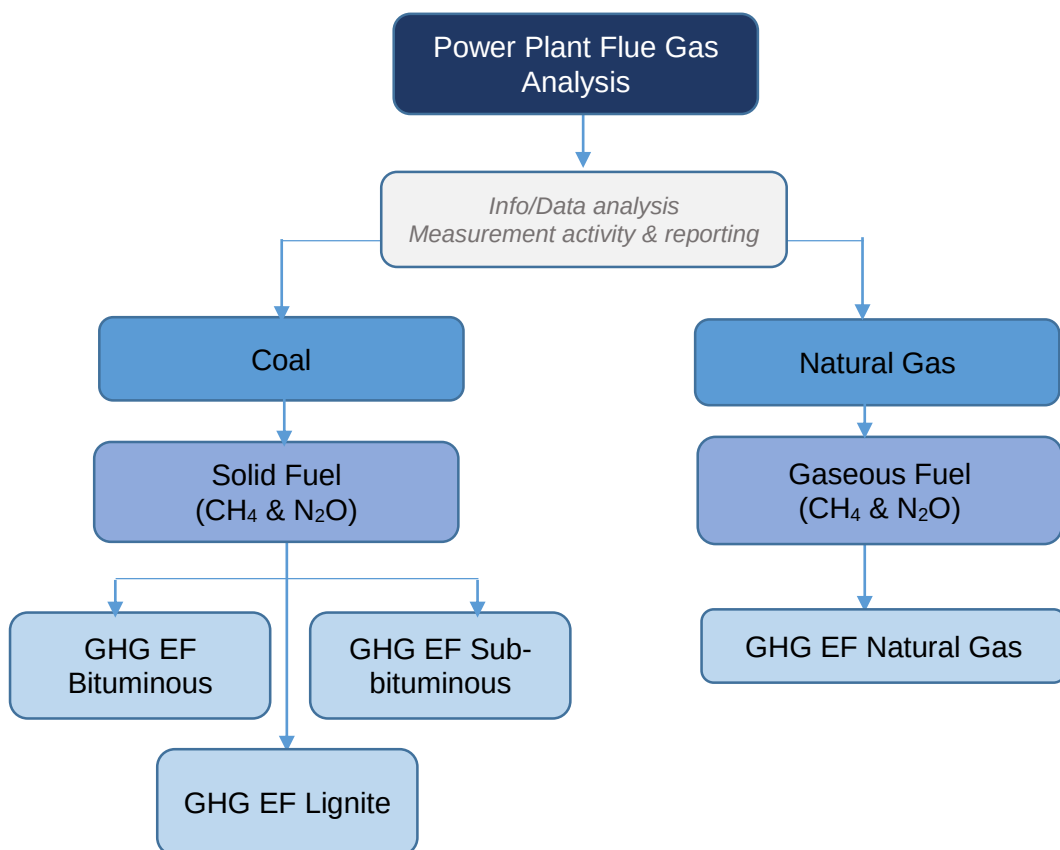


Figure 3: CH₄ and N₂O Emission Factor Framework

The required data to perform non-CO₂ emission factor calculations are listed in the following **Table 1**:

Table 1: Required Data and Methods for Obtaining Emission Factor

Data Required	Purpose	Method	Source
Type of fuel	Classification of emission factors	Operator information	Power plant supplied data
Elemental analysis values of fuel	Calculation of combustion emissions	COA / lab test	
Net calorific value of fuel	Calculation of emission factor denominator	COA / lab test	
CH ₄ concentration	CH ₄ emission calculation	Equipment measurement	Actual measurement data
N ₂ O concentration	N ₂ O emission calculation	Equipment measurement	
O ₂ concentration	Excess air ratio calculation	Equipment measurement	
Combustion exhaust emissions	CH ₄ & N ₂ O emission calculation	Calculation	Calculated data

Emission factor calculations based on direct measured emissions for both coal and natural gas utilise the same equations. These equations are sourced from the National Greenhouse Gas Inventory Report of Japan (Apr. 2015).

The emission factor from flue gas analysis are obtained by calculating volumetric flow rate value as shown in below Equation:

$$\begin{aligned}
 EF_{CH_4} &= C_{CH_4} \times [G_0 + (m - 1) A_0] \text{ mw}_{CH_4} / V_m / NCV \\
 EF_{N_2O} &= C_{N_2O} \times [G_0 + (m - 1) A_0] \text{ mw}_{N_2O} / V_m / NCV \\
 m &= 21 / (21 - C_{O_2})
 \end{aligned}$$

Symbols used and along with their definitions and units are listed as follow:

EF_{CH_4 / N_2O} : CH₄ or N₂O emission factor, [kg/TJ];
 C_{CH_4 / N_2O} : CH₄ or N₂O concentration in exhaust gas, [ppm];

- G_0 : theoretical exhaust gas volume for each unit mass of fuel combustion (dry), [m^3/kg];
- A_0 : theoretical air volume for each unit mass of fuel combustion, [m^3/kg];
- M : air ratio (actual air volume divided by theoretical air volume), [-];
- $\text{mw}_{\text{CH}_4/\text{N}_2\text{O}}$: molecular weight of measured gas (constant at 16 g/mol for CH_4 or 44 g/mol for N_2O), [g/mol];
- V_m : one mole ideal gas volume in standardized condition (constant at $22.4 \times 10^{-3} \text{ m}^3/\text{mol}$), [m^3/mol];
- NCV : net calorific value for each fuel combustion, [MJ/kg]; and
- C_{O_2} : O_2 concentration in exhaust gas, [%]



Figure 4: Plant stack emissions measurement for Non- CO_2 emission factor

3 DATA VALIDATION AND VERIFICATION

- **Validation**

Validation was also conducted for the coal and natural gas composition. TNBR Fuel and Combustion Unit was engaged to sample and analyse the fuel gas supplied to selected natural gas power plants for validation purposes. This activity was conducted to check the accuracy and quality of source data before being used in the analysis of country specific emission factor. It is crucial that the collected data be evaluated and validated for the proper interpretation of the large data set.

The test programme for coal samples consisted of analytical testing activities which involved sample preparation, calorific value determination, proximate analysis and ultimate analysis. These were carried out by TNBR Fuels and Combustion Unit. The testing procedures were carried out according to various ASTM procedures. For the natural gas, the sampling programme was carried out using an Agilent 490 Micro GC natural gas analyser that was designed to identify different compositions of natural gas. Based on this activity, no significant difference between data supplied by data provider (i.e COA & Daily Gas Report) with verification from TNBR Fuels and Combustion Unit.

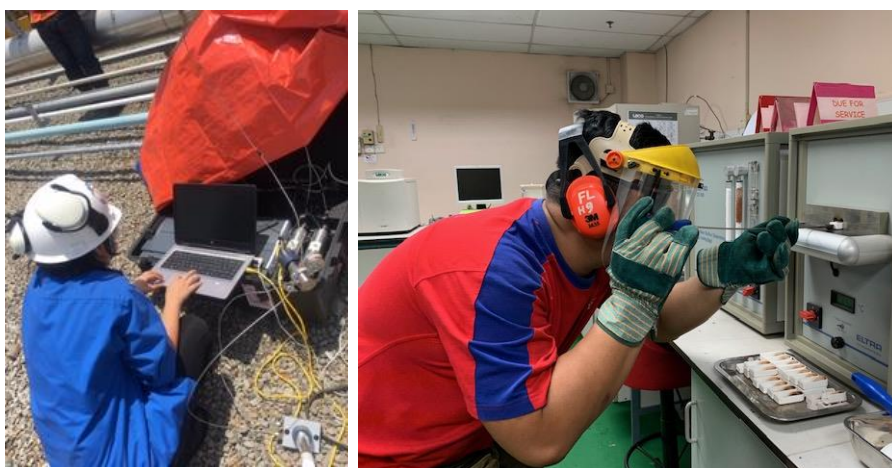


Figure 5: Fuel composition verification (Natural Gas & Coal)

- **Verification**

Verification is an assessment to check the accuracy and completeness of GHG reporting information. The study focused on the information that is relevant, complete, accurate, consistent, transparent, and without material misstatements. The verification process demonstrates a level of assurance or confidence in the findings. It can also be related to quality assurance and quality control. This study used a combination of verification (ISO 14064-3:2019 - Greenhouse gases — Part 3: Specification with guidance for the verification and validation of greenhouse gas statements) and QA/QC (IPCC2006, Vol. 1, Chap. 6 - QA/QC and Verification) to conduct the data verification process.

The main activity in the verification was to check the primary sources that was used in emission reporting and to carry out the observations on the completeness of the data set. Next, test and controls for potential error checking was been conducted. It covers error in input, transformation and output categories as tabulated in **Table 2**.

Table 2: Error Checking and Methodologies

Error checking categories	Test and control
Input	• Record count • Valid character test • Missing data test
Transformation	• Blank test (9% from dataset) • Consistency test • Cross-checking test • File control
Output	• Statistical test • Multivariable analysis

In ensuring the QA/QC of the project are met, the cataloguing and verification process was conducted which involved thorough checks on data health – pertaining consistency of data, existence of data, formula and format. The data cataloguing process is an important step in developing dataset (data collection) to ensure a globally acceptable level which is comprised of

metadata with objects such as base tables, synonyms or indexes. Cataloguing data is important, and it allows greater accuracy and time for data movement and presentation in project deployments. Additionally, extra data governance can be imposed towards achieving higher data consistency. Data can be organised into two main groups: (1) primary dataset; and, (2) secondary dataset. For this project there are three primary datasets:

- i. Certificate of Analysis / Certificate of Arrival; – coal fuel data.
- ii. Gas chromatography analysis results – for natural gas.
- iii. Flue gas measurements (Gasmeter® DX4000 flue gas analyzer) – for non-CO₂ emissions for all plants.

- **Expert Peer Review and Verification**

TNBR also conducted the verification with local and international experts for the emission factor developed. The experts who were engaged in this exercise are:

- i. **Professor Dr. Nik Meriam Nik Sulaiman**

Department of Chemical Engineering
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- ii. **Profesor Dr. Jeon Eui-Chan**

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Review and check the monitoring data and verify that the monitoring methodologies applied for the measurement and estimation of anthropogenic emissions for the sources have been applied correctly, and the documentation is complete and transparent.

This verification exercise's objective was to confirm that the country-specific GHG emission factor as documented within the provided documentation was accurate, complete, consistent, transparent, and free of material error or omission in accordance with the criteria outlined herein.

Based on the data and information provided by TNBR for this exercise, it was conclude with assurance within the materiality threshold of 5% for aggregate errors, and there is no evidence that the country-specific GHG emissions factor reported and emission factor derived:

- is not materially correct;
- is not a fair representation of the country-specific GHG emissions data and information; and
- is not prepared in accordance with the criteria listed above.

4 UNCERTAINTY ASSESSMENT

Uncertainty is used to characterize the dispersion of values that could be reasonably attributed to a measured quantity (IPCC, 2000). Uncertainty information is not intended to dispute the validity of the inventory estimates, but to help prioritise efforts to improve the accuracy of emission assessment in the future and guide decisions on methodological choice.

Uncertainty of the figures produced by this study is tracked and reported on all major sources of uncertainty from the equipment, measurement, and process. The general steps for quantifying uncertainty are:

- i. Determine uncertainty for emission factor for every power plant according to year (2017-2019);
- ii. Determine combined uncertainties.

Quantifying Uncertainty

In terms of the emission factor, the standard deviation for the calculated factor averages can be determined using below equation:

$$\text{Power plant EF } \sigma_{\text{CO}_2 / \text{CH}_4 / \text{N}_2\text{O}} = \left[\sum (\text{EF}_i - \text{EF}_{\text{ave, plant}})^2 / (N - 1) \right]^{1/2}$$

Where,

$\text{EF}_{\text{CO}_2 / \text{CH}_4 / \text{N}_2\text{O}}$	: uncertainty of CO ₂ , CH ₄ , or N ₂ O emission factor for data obtained from power plant, [kg/TJ]
EF_i	: emission factor for the combustion fuel at data point i, calculated per shipment or daily or monthly, [kg/TJ];
$\text{EF}_{\text{ave, plant}}$	: average emission factor of the power plant for the entire set of i data points, [kg/TJ];
N	: total number of data points for the power plant

The difference between EF_i and EF_{ave} is called variance and is calculated for every data point included in the emission data population.

According to 2006 IPCC Guidelines, good practice requires the use of a 95 percent confidence interval for quantification of random errors. This may also be expressed as a percentage of the central estimate. Assuming a uniform probability density function, the confidence interval can be conveniently expressed as plus or minus half the confidence interval width divided by the estimated value of the variable (e.g., $\pm 10\%$). For normal distributions the 95 percent confidence interval would be plus or minus twice the estimated standard deviation of the population, as described in below equation:

$$U_{i \text{ CO}_2/\text{CH}_4/\text{N}_2\text{O}} = 2 \sigma_{\text{CO}_2/\text{CH}_4/\text{N}_2\text{O}}$$

The uncertainty U_i is then used to arrive at the overall uncertainty in national emissions and possible trends in national emissions in the following section.

Methods to Combine Uncertainties

Combination of uncertainties is done when values that are contained in them uncertainties are used to calculate other values. For addition or summation operations involving values that have uncertainties attached, the uncertainties themselves can be resolved by below Equation:

$$U_{\text{total}} = [U_1^2 + U_2^2 + U_i^2 + \dots + U_n^2]^{1/2}$$

Similarly, multiplication operations of uncertainties are calculated in below Equation with the use of multiplication factors:

$$U_{\text{total}} = \frac{[(U_1 x_1)^2 + (U_2 x_2)^2 + (U_i x_i)^2 + \dots + (U_n x_n)^2]^{1/2}}{x_1 + x_2 + x_i + \dots + x_n}$$

Where;

- U_{total} : percentage uncertainty in the sum or product of the quantities half the 95 percent confidence interval divided by the total and expressed as a percentage), [unit of the calculated quantity];
- $U_{i, n}$: uncertain quantity being calculated, [unit of the calculated quantity]; and
- $x_{i, n}$: percentage of uncertainties associated with the final quantity, [-]

Considering x_i as the weight values it can be used to calculate the uncertainty for the national greenhouse gas emission factor.

Comparison to IPCC 2006 default values shows that data presented are reliable and valid within the acceptable level of confidence. Based on the data and information collected and the processes and procedures conducted, the assurance are within the materiality threshold of 5% for aggregate errors and there is no evidence that the country-specific GHG emissions factor reported and emission factor derived:

- is not materially correct;
- is not a fair representation of the country-specific GHG emissions data and information; and
- is not prepared in accordance with the criteria listed above.

5 KEY RECOMMENDATIONS

To improve overall data accuracy and consistency, it is necessary to develop a continuity in the process and procedure at the national level to manage the uncertainties and discrepancies embedded in the electricity sector data. Recommendations for future works are presented in this section.

i. Adoption of country-specific emission factor for coal and natural gas

The developed country-specific emission factor are proposed to be adopted for the national greenhouse gas emissions inventory assessment process. This will help to improve the accuracy level of GHG emissions reported. Additionally, submission to IPCC Emission Factor Database System (IPCC EFDB) is strongly suggested to ensure the recognising Malaysia's effort in GHG emissions inventory activities.

ii. Establishment of framework for future Monitoring, Reporting and Verification (MRV).

At national level, the implementation of an MRV framework covers two main areas: Development of a domestic framework for MRV, based on guidelines for MRV of domestically supported NAMAs; and, Implementation of the international MRV requirements, such as measurement and reporting through the national communications and BURs. As such, the developed National GHG Emissions Inventory System shall serve as an important platform to be referred as one of the tools in supporting the MRV for electricity sector.

iii. Establishment of Government Mechanism for Reporting of Greenhouse Gases Programme

Government support is crucial for the success of a national programme. The government can establish a national registry for the annual reporting of greenhouse gases for the power plants by introducing a Reporting of Greenhouse Gases Programme. This programme will support the efforts in preparing the National Communication which have to be submitted to the UNFCCC.

iv. Continual Periodic Updates of Emission Factor and Reporting to the National GHG Emissions Inventory Database System

There is a need to ensure the continued updating of the coal and natural gas emission factor in order to maintain the GHG emission factor and calculate a credible emission factor. This is because the calorific value and the amounts of fuel composition are not fixed parameters and are dependent on fuel characteristics. Additionally, the GHG emissions reporting and monitoring by the developed system are readily adapted by the power plants. The main reasons for having a National GHG Emissions Inventory Database System are to maintain and organise data for daily use, for future improvement of GHG Inventory, and as a reference database. The programme also allows power plants to record their reductions and ensure that they are not penalised under a future climate policy.

v. Verification of GHG Emission Factor and GHG Emissions

Third-party verification of emission factor is required for reporting to produce an accurate and equitable apportionment of emission factor and emissions. A detailed verification and validation standard needs to be set up or adopted for this purpose.

vi. Study on Other Fuel Type Emission Factors and Development of National Electricity System Emission Factor

An extension to other fuels by using the same methodology and approach is recommended. With this, a complete country-specific emission factor for the electricity sector can be established. In addition, by referring to the established guidelines, the establishment of other sector of country specific emission factor can be explored. Furthermore, the development of a national electricity system emission factor and impact assessment is a very important aspect towards aligning with future MESI 2.0 whereby a grid connected fuel type including RE will be the main sources. This project is important to fulfil the requirement for the grid of the future whereby it quantifies and set the reference value on national emission factor according to the fuel source and area for the grid electricity system in Malaysia.

vi. Life Cycle Assessment

Life Cycle Assessment (LCA) is required to achieve more comprehensive assessment of a power plant operations. It takes into consideration the assessment of various materials used during the design, construction and other stages involved in the electricity sector. LCA is useful

for power plants during the planning and construction of new "green" power plants/substations/power lines, in deciding which materials to use, or which fuel type to consider. Additionally, it will support other applications of emission factor such as product carbon foot printing and environmental impact assessment.

ACKNOWLEDGEMENT

We would like to express our deepest gratitude to Akaun Amanah Industri Bekalan Elektrik (AAIBE), Ministry of Energy, Science, Technology, Environment and Climate Change and TNB Research Sdn. Bhd. for providing the funding and continuous support in completing this project on "Development Of Greenhouse Gas (GHG) Emission Factor For The Electricity Sector In Malaysia". Our high appreciation goes to the project team members all relevant offices and everyone who directly or indirectly has contributed towards the successful completion of this project.

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